

Principles of Numerical Control and Coding

Automation, NC / CNC / DNC, Part Programming, APT & Economics

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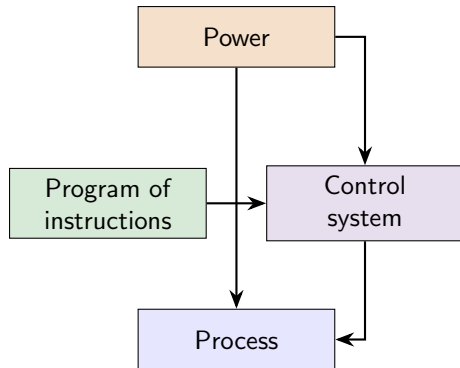
- 1 Basic Principles of Automation
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What is Automation?

Definition

Automation is the technology by which a process or procedure is carried out with **minimum human assistance**, using a **program of instructions** combined with a **control system** that executes those instructions.

- Every automated system has **three** basic elements:
 - ① **Power** to run the process and the controls.
 - ② A **program of instructions** (what to do, in which order).
 - ③ A **control system** that carries out the program.
- In a machine tool the “program” is the *part program* and the control system is the *NC controller*.



Types of Automation

Fixed

Hard automation

- Sequence fixed by the equipment.
- Very high volume, one product.
- Low cost per part.
- Transfer lines.

Programmable

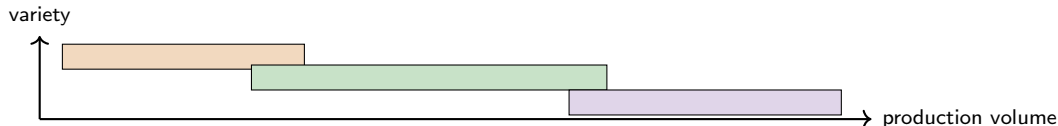
Change the program

- New product = new program.
- Batch production, medium volume.
- Set-up time lost.
- **NC / CNC**, PLCs.

Flexible

No lost changeover time

- Mix of different parts run together.
- Set-ups prepared off-line.
- Flexible mfg. systems (FMS).



Reasons for Automating and the USA Principle

Why automate?

- Increase labour **productivity**.
- Reduce labour cost & overcome labour shortages.
- Remove workers from **unsafe** or tedious tasks.
- Improve **quality** and consistency.
- Reduce manufacturing **lead time**.
- Do what cannot be done manually (complex 3-D shapes).

The USA principle

Before automating any process, follow three steps:

- 1 **Understand** the existing process.
- 2 **Simplify** it — remove unnecessary steps.
- 3 **Automate** what remains.

Automating a bad process only makes bad parts *faster*.

Ten Strategies for Automation and Process Improvement

#	Strategy	Example
1	Specialisation of operations	special-purpose tools for one operation
2	Combined operations	several cuts in one set-up (machining centre)
3	Simultaneous operations	several tools cutting at the same time
4	Integration of operations	machines linked by automatic transfer
5	Increased flexibility	same machine for many parts (CNC)
6	Improved material handling	conveyors, AGVs, automated storage
7	On-line inspection	probing the part on the machine
8	Process control & optimisation	adaptive control of feed and speed
9	Plant operations control	computer networks, DNC
10	Computer-integrated manufacturing (CIM)	CAD, CAM, planning in one database

Strategies 2, 5, 7, 8 and 9 are exactly what **NC, CNC, DNC and adaptive control** provide.

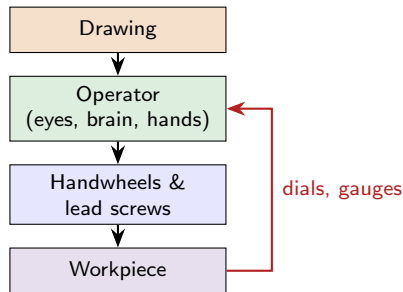
Limitations of Conventional Machine Tools

A conventional lathe or milling machine

The operator reads the drawing, turns the handwheels, watches the dials and decides every movement.

- Accuracy depends on operator **skill** and fatigue.
- Complex contours are slow or impossible.
- Long set-up and idle (non-cutting) time.
- Poor repeatability from part to part.

The human is the “control system” — the first step is to give the machine devices that take over part of that job.



Improved Devices on Conventional Machines

Device	What it does	Benefit
Power feeds & rapid traverse	motor drives the slides instead of the operator	less effort, uniform feed
Digital read-out (DRO)	linear scales display slide position (0.005 mm)	no dial errors or backlash
Copying / tracer attachment	a stylus follows a template; hydraulics copy it	repeated contours
Turret / capstan head	several tools pre-set in one indexing head	fast tool change
Cam-operated automatics	cams sequence every movement	fully automatic, high volume
Jigs, fixtures, quick-change tool posts	locate the part / tool in seconds	short set-up

Key idea

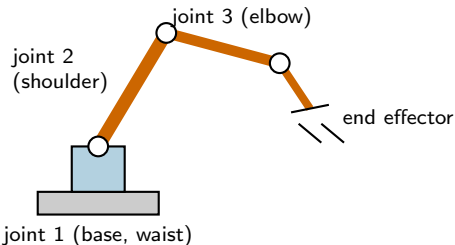
Each device automates *one* part of the operator's job. **Numerical control** automates *all* of it — the movements come from a program instead of cams or templates.

Manipulators and Industrial Robots

Manipulator

A mechanical arm that moves parts or tools. A **pick-and-place** unit follows a fixed sequence; an **industrial robot** is a *programmable*, multi-axis manipulator.

- Load / unload machine tools and presses.
- Change tools and pallets; transfer parts between machines.
- Welding, painting, assembly, inspection.
- One robot can serve several CNC machines (a **cell**).



Articulated arm: 3 arm joints position the wrist; the wrist (2–3 joints) orients the gripper.

Robot Configurations and End Effectors

Configuration	Arm joints	Typical use
Cartesian (gantry)	L L L (three linear)	pick-and-place, CNC loading
Cylindrical	R L L (rotate, lift, extend)	machine loading
Polar (spherical)	R R L	spot welding, die casting
SCARA	R R L (vertical axes)	fast electronic assembly
Articulated (jointed arm)	R R R	welding, painting, general

Grippers

Mechanical fingers, vacuum cups, magnets — they *hold* the part.

Tools

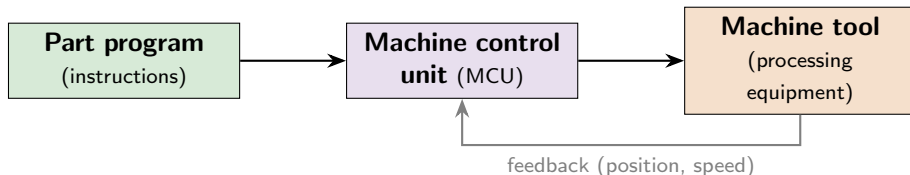
Welding torch, spray gun, spindle, screwdriver — they *work on* the part.

L = linear (sliding) joint, R = rotary joint.

What is Numerical Control (NC)?

Numerical Control

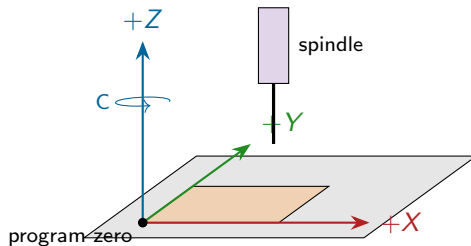
A form of **programmable automation** in which the mechanical actions of a machine tool are controlled by a **program** of coded **numbers, letters and symbols**.



- **Part program** — step-by-step instructions: tool positions, feeds, speeds, tool changes, coolant.
- **MCU** — reads and decodes the program and drives the axes (today: a computer → **CNC**).
- **Machine tool** — table, spindle, slides, motors that actually cut the metal.
- History: J. T. Parsons & MIT, first NC milling machine in **1952** (US Air Force parts).

The NC Coordinate System

- Uses the **right-hand** Cartesian system (ISO 841).
- **Z axis** is always along the **spindle**; +Z moves the tool *away* from the work.
- **X axis** is the longest horizontal travel; **Y** completes the right-hand set.
- Rotary axes **A, B, C** rotate about X, Y, Z (5-axis machines).
- Positions are measured from the **program zero** (work datum) chosen by the programmer.



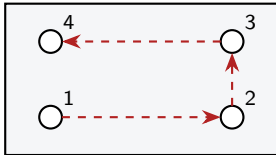
Vertical milling machine: tool moves in Z, table moves in X and Y.

Point-to-Point vs. Continuous-Path (Contouring) Control

Point-to-point (PTP)

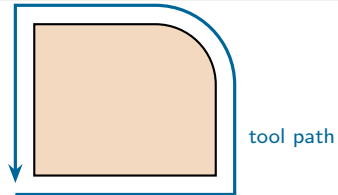
Tool moves to a position **without cutting**, then the operation is done there. The path between points does not matter.

Drilling, punching, spot welding.



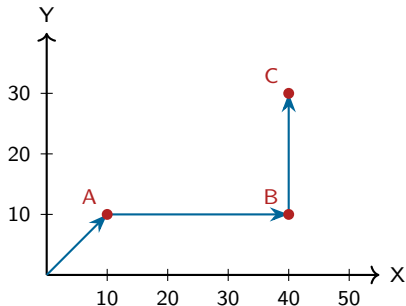
Continuous path (contouring)

All axes move **simultaneously** under control, so the tool follows any line or curve **while cutting**.
Milling profiles, turning tapers & radii.



Contouring needs **interpolation**: the MCU calculates the intermediate points between the programmed end points.

Absolute vs. Incremental Positioning



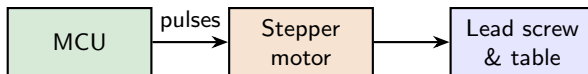
Move	Absolute (G90)	Incremental (G91)
0 → A	X10 Y10	X10 Y10
A → B	X40 Y10	X30 Y0
B → C	X40 Y30	X0 Y20

- **Absolute:** every position measured from program zero — an error does not carry forward.
- **Incremental:** each move measured from the *last* position — handy for repeated patterns.

Open-Loop and Closed-Loop NC Systems

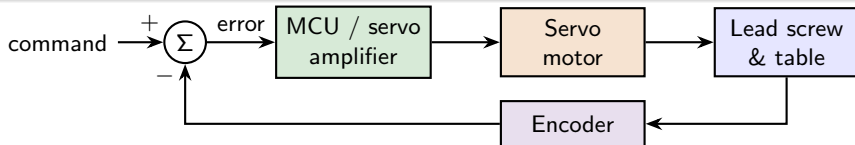
Open loop

The MCU sends a number of **pulses** to a **stepper motor**; it assumes the table got there. Cheap, but no correction if steps are missed.



Closed loop

A **servo motor** drives the table and an **encoder** or linear scale feeds the actual position back; the MCU corrects any error.



Basic Length Unit (BLU) — Worked Example

Basic Length Unit

The **smallest** distance the table can be moved by one pulse — the **control resolution** of the axis.

$$\text{BLU} = \frac{p}{n_s r_g} \quad f_p = \frac{v r_g n_s}{60 p}$$

p = lead-screw pitch (mm), n_s = steps/rev, r_g = gear ratio (motor : screw), v = feed (mm/min).

Problem: open-loop axis, pitch $p = 6$ mm, stepper 200 steps/rev, gear ratio 5:1. Find the BLU, the pulses for a 250 mm move, and the pulse rate for a feed of 500 mm/min.

$$\text{BLU} = \frac{6}{200 \times 5} = \boxed{0.006 \text{ mm}}$$

$$n_p = \frac{250}{0.006} \approx \boxed{41\,667 \text{ pulses}}$$

$$f_p = \frac{500 \times 5 \times 200}{60 \times 6} \approx \boxed{1389 \text{ Hz}}$$

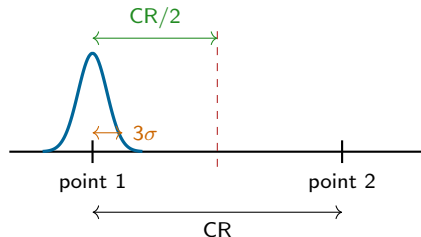
Precision: Resolution, Accuracy and Repeatability

- **Control resolution (CR)** — distance between two adjacent addressable points (= BLU).
- **Accuracy** — worst-case error in reaching a target:

$$\text{Accuracy} = \frac{\text{CR}}{2} + 3\sigma$$

- **Repeatability** — ability to return to the *same* point: $\pm 3\sigma$.

σ = standard deviation of the mechanical errors (backlash, deflection, thermal).



Example: CR = 0.006 mm, $\sigma = 0.002$ mm

$$\text{Accuracy} = 0.003 + 0.006 = \boxed{0.009 \text{ mm}}$$

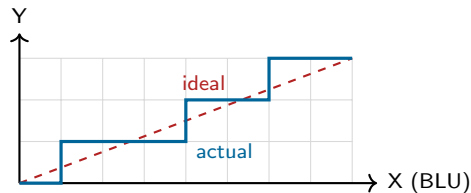
$$\text{Repeatability} = \pm 0.006 \text{ mm}$$

Interpolation: Moving Along a Line or Arc

Interpolator

The part program gives only the **end points**; the interpolator in the MCU generates the pulse sequence for each axis so the tool stays close to the ideal path.

- **Linear** (G01) — straight line in 2 or 3 axes.
- **Circular** (G02 CW / G03 CCW) — arc in one plane.
- Helical, parabolic, spline — on advanced controls.
- The path is really a **staircase** of BLU steps — invisible because a BLU is only a few μm .



Line from (0,0) to (8,3) BLU — the C program later in this deck generates exactly these steps.

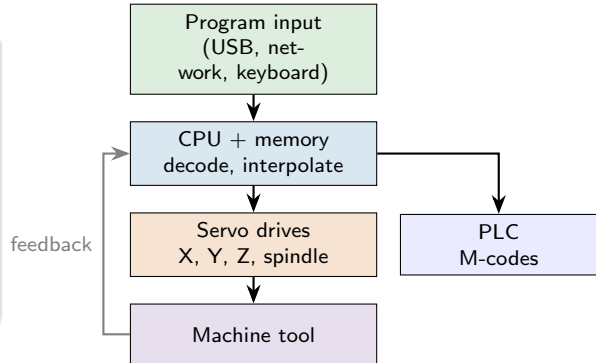
Computer Numerical Control (CNC)

CNC

An NC system whose MCU is a **dedicated computer**. The part program is stored in memory and executed by software.

Features of CNC

- Program stored, edited & optimised at the machine.
- Interpolation done in software (any type).
- Tool-length and cutter-radius compensation.
- Canned cycles, sub-programs, macros.
- Graphic simulation, diagnostics, communication.

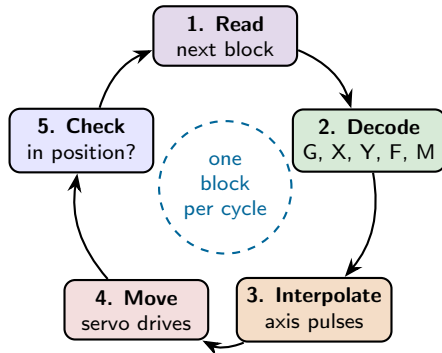


How a CNC Executes a Part Program: The Block Cycle

The Read—Decode—Interpolate—Move cycle

Just like the CPU's machine cycle, the CNC repeats one cycle for every block (line) of the part program until it reads M30.

- 1 **Read** — fetch the next block, e.g. N60 G01 X60 F200.
- 2 **Decode** — split it into words: motion type, target, feed, M-codes.
- 3 **Interpolate** — compute the intermediate points and pulse rates for each axis.
- 4 **Move** — servo drives move the axes; the PLC handles spindle, coolant, tool change.
- 5 **Check** — feedback confirms position is reached “in position”, then go to the next block.

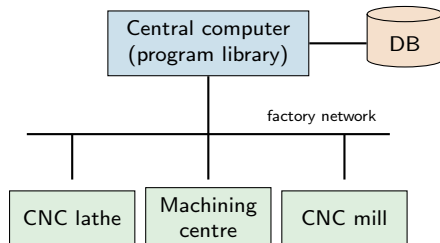


Direct Numerical Control (DNC)

DNC

A **central computer** stores the part programs and sends them to many machine tools over a network — originally “direct”, today **distributed** NC.

- One library of proven programs — no lost or outdated tapes.
- **Drip-feeding**: very long 3-D programs streamed block by block.
- Collects production data: run time, part count, alarms.
- Basis of **FMS** and computer-integrated manufacturing.



Comparing NC, CNC and DNC

Feature	NC	CNC	DNC
Controller	hard-wired logic	dedicated computer	central computer + CNCs
Program medium	punched tape, read every part	stored in memory	stored on server, sent over network
Editing	punch a new tape	at the machine	at the central computer
Interpolation	hardware	software	software (in each CNC)
Machines	one	one	many
Data feedback	none	diagnostics on screen	production reports

Key idea

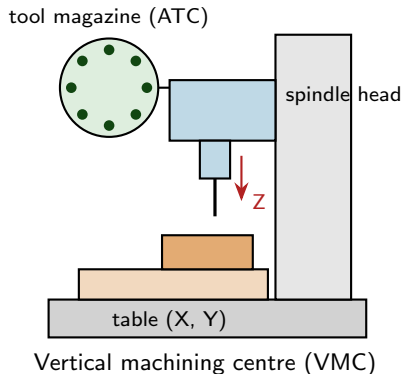
NC → CNC put a computer *in* the machine; CNC → DNC connects the machines *to each other*.

Machining Centres

Machining centre

A CNC machine tool that performs **milling, drilling, boring, tapping** on several faces of a part in **one set-up**, with an **automatic tool changer**.

- **ATC** — tool magazine (20–120 tools) and a change arm; tool change in 1–5 s.
- **APC** — automatic pallet changer: load the next part while cutting.
- **Types**: vertical (VMC), horizontal (HMC), universal / 5-axis.
- **Turning centre** — the CNC lathe equivalent, with live tools.



How NC Programs are Stored and Coded

Program input media — then & now

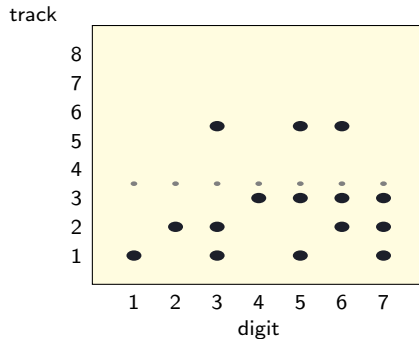
- **Punched tape** (1 inch, 8 tracks) — the classic NC medium.
- Magnetic tape, floppy disk.
- Keyboard entry at the CNC (MDI).
- USB stick, memory card, Ethernet / DNC.

Two standard character codes

EIA RS-244 — odd parity, track 5 is the parity track.

ISO / ASCII (RS-358) — even parity, track 8 is the parity track.

EIA code for the digits 1–7



Tracks 1–4 carry the BCD weights 1, 2, 4, 8. Digit 3 (1+2) has two holes, so track 5 adds a parity hole to make the count **odd**.

Block Formats: Coding an NC Instruction

Block

One line of an NC program = one complete instruction (a move plus its conditions).

Format	Rule	Example
Fixed sequential	every word, fixed order & length, no letters	0050 01 02500 03000 200
Tab sequential	fixed order, words separated by TAB; repeated words omitted	050 → 01 → 25.0 → 30.0
Word address	each word starts with a letter ; any order, only changed words	N050 G01 X25.0 Y30.0 F200 S1000 M03

Today

Every modern CNC uses the **word-address format** (ISO 6983 “G-code”). The letter tells the controller what the number means.

Words in a Word-Address Block

Letter	Meaning	Example
N	sequence (block) number	N050
G	preparatory function (type of motion / mode)	G01 linear feed
X Y Z	coordinates of the end point	X25.0 Y30.0
I J K / R	arc centre offset / radius	R10.0
F	feed rate (mm/min)	F200
S	spindle speed (rpm)	S1000
T	tool number	T02
M	miscellaneous function (on/off)	M03 spindle CW

Read a block like a sentence: N050 G01 X25.0 Y30.0 F200 → “step 50: feed in a straight line to (25, 30) at 200 mm/min”.

Methods of NC Part Programming

Manual

G-code written by hand

- Programmer calculates every coordinate.
- Best for simple 2-D parts, PTP drilling.
- Error-prone for curves.

Computer-assisted

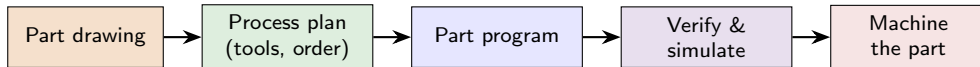
APT and similar languages

- Describe geometry & tool motion in English-like words.
- Computer does the maths.
- Post-processor makes G-code.

CAD/CAM & conversational

Graphical

- Tool paths generated from the 3-D CAD model.
- Simulation before cutting.
- Conversational: fill-in screens at the CNC.



Common G-Codes (Preparatory Functions)

Code	Function	Code	Function
G00	rapid positioning (no cutting)	G40	cancel cutter-radius compensation
G01	linear interpolation (feed)	G41	cutter compensation, tool left
G02	circular interpolation, CW	G42	cutter compensation, tool right
G03	circular interpolation, CCW	G43	tool-length compensation
G04	dwell (pause)	G80	cancel canned cycle
G17	XY plane selection	G81	drilling canned cycle
G20 / G21	inch / metric units	G90	absolute programming
G28	return to reference point	G91	incremental programming

Modal codes

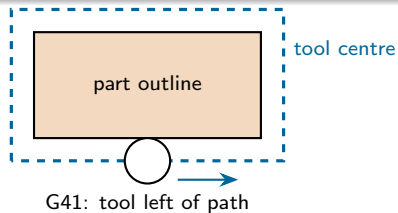
Most G-codes are **modal**: they stay active until replaced by another code of the same group — so G01 need not be repeated on every line.

Common M-Codes and Cutter-Radius Compensation

Code	Function
M00	program stop
M02	end of program
M03 / M04	spindle ON, CW / CCW
M05	spindle stop
M06	tool change
M08 / M09	coolant ON / OFF
M30	end of program & rewind

Cutter-radius compensation

The programmer writes the **part outline**; the CNC offsets the tool centre by the cutter radius.

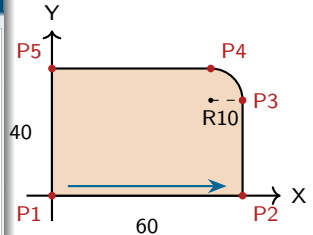


Machining plan

- 1 START; metric, absolute, XY plane
- 2 Load tool 1 ($\varnothing 10$ end mill)
- 3 Rapid above P1, spindle & coolant ON
- 4 Feed down to $Z = -2$
- 5 Cut P1 $\rightarrow$ P2 $\rightarrow$ P3
- 6 Arc P3 $\rightarrow$ P4 (R10, CCW)
- 7 Cut P4 $\rightarrow$ P5 $\rightarrow$ P1
- 8 Retract, coolant & spindle OFF
- 9 STOP (M30)

Manual part program (G-code)

```
O1001 (PROFILE MILLING)
N10 G21 G90 G17
N20 T01 M06 (10 MM END MILL)
N30 G00 X0 Y0 Z5.0
N40 S1200 M03 M08
N50 G01 Z-2.0 F100 (PLUNGE)
N60 X60.0 F200 (P1 TO P2)
N70 Y30.0 (P2 TO P3)
N80 G03 X50.0 Y40.0 R10.0
N90 G01 X0 (P4 TO P5)
N100 Y0 (P5 TO P1)
N110 G00 Z50.0 M09
N120 M05
N130 M30
```

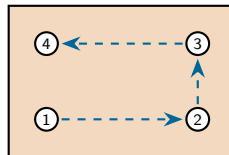


Program zero at P1 (lower-left corner), top of part = Z0. Tool centre follows the outline (no compensation) for clarity.

Canned Cycles: Drilling with G81

```
N10 G21 G90 G17
N20 T02 M06          (8 MM DRILL)
N30 S900 M03 M08
N40 G00 X10.0 Y10.0 Z5.0
N50 G81 Z-15.0 R2.0 F80
N60 X50.0           (HOLE 2)
N70 Y30.0           (HOLE 3)
N80 X10.0           (HOLE 4)
N90 G80 G00 Z50.0 M09
N100 M05
N110 M30
```

- A **canned cycle** packs a repeated sequence into one code.
- G81 Z-15 R2 F80 means: rapid to $R = 2$, feed down to $Z = -15$, rapid back up — at every new X/Y.
- Four holes need only **one** cycle line plus the hole positions.
- G80 cancels the cycle.



Pseudocode

- 1 START
- 2 READ end point dx, dy (BLU)
- 3 $err = 2dy - dx$
- 4 FOR each of the dx steps
 - pulse X by 1 BLU
 - if $err > 0$: pulse Y, $err -= 2dx$
 - $err += 2dy$
 - PRINT X, Y
- 5 STOP

C Program (linear interpolator)

```
#include <stdio.h>

int main(void)
{
    int dx, dy, x = 0, y = 0, err, i;

    printf("End point in BLU (x y): ");
    scanf("%d %d", &dx, &dy); /* dx>=dy>=0 */

    err = 2*dy - dx;          /* decision */
    for (i = 1; i <= dx; i++) {
        x++;                  /* pulse X */
        if (err > 0) {
            y++;              /* pulse Y */
            err -= 2*dx;
        }
        err += 2*dy;
        printf("Step %d : X=%d Y=%d\n",
               i, x, y);
    }
    return 0;
}
```

Program Output

End point in BLU (x y):

8 3

Step 1 : X=1 Y=0

Step 2 : X=2 Y=1

Step 3 : X=3 Y=1

Step 4 : X=4 Y=1

Step 5 : X=5 Y=2

Step 6 : X=6 Y=2

Step 7 : X=7 Y=3

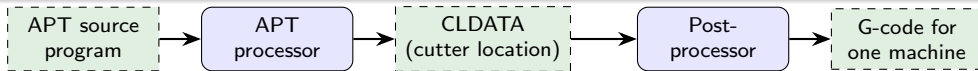
Step 8 : X=8 Y=3

Note: these are the steps drawn on the *Interpolation* slide. A real CNC runs this logic thousands of times per second for every axis.

APT: Automatically Programmed Tools

APT

The first and most widely used **NC part-programming language**, developed at MIT (1956–59). The programmer describes the **geometry** and the **tool motion** in English-like statements; the computer calculates the tool path.



- The **processor** is machine-independent: it computes the cutter-centre path (CLDATA).
- The **post-processor** converts CLDATA into the codes of a *particular* machine and controller.
- Same idea as a C compiler + linker: one source program, many target machines.

The Four Types of APT Statements

Type	Purpose	Examples
Geometry	define points, lines, circles, planes	P1 = POINT/0, 0, 0 L1 = LINE/P1, P2 C1 = CIRCLE/CENTER, PC, RADIUS, 10
Motion	describe the tool path	FROM/P0 GOTO/P1 GORGT/L1, PAST, L2
Post-processor	speeds, feeds, coolant, machine	SPINDL/1200, CLW FEDRAT/200 COOLNT/ON
Auxiliary	part name, cutter size, tolerance, end	PARTNO CUTTER/10 TOLER/0.01 FINI

Each statement: MAJOR WORD / minor words, values. Comments start with \$\$.

APT Motion Statements

Point-to-point

FROM/P0 — starting point

GOTO/P1 — go to an absolute point

GODLTA/20, 0, -5 — incremental move

Contouring — direction words

GOLFT / GORGT — turn left / right

GOFWD / GOBACK — go forward / back

GOUP / GODOWN — up / down

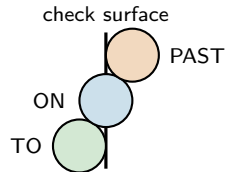
Where the tool stops (check surface)

TO — tool just touches it

ON — tool centre on it

PAST — tool just beyond it

TANTO — tangent point of a curve



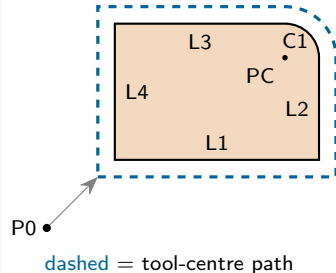
Tool position relative to the drive surface: TLLFT, TLRGT (left / right of it) or TLON (on it).

APT (1): set-up & geometry APT (2): motion

```
PARTNO PROFILE MILLING
MACHIN/MILL, 1
UNITS/MM
CUTTER/10.0
TOLER/0.01
$$ ----- geometry -----
P0 = POINT/-20, -20, 10
P1 = POINT/0, 0, 0
P2 = POINT/60, 0, 0
P3 = POINT/60, 30, 0
P5 = POINT/50, 40, 0
P6 = POINT/0, 40, 0
PC = POINT/50, 30, 0
L1 = LINE/P1, P2
L2 = LINE/P2, P3
C1 = CIRCLE/CENTER, PC, $
    RADIUS, 10
L3 = LINE/P5, P6
L4 = LINE/P6, P1
PL1 = PLANE/0, 0, 1, -2
```

```
$$ ----- machining -----
SPINDL/1200, CLW
FEDRAT/200
COOLNT/ON
FROM/P0
GO/T0, L1, T0, PL1, T0, L4
TLRGT, GORGT/L1, PAST, L2
GOLFT/L2, TANTO, C1
GOFWD/C1, TANTO, L3
GOFWD/L3, PAST, L4
GOLFT/L4, PAST, L1
GOTO/P0
COOLNT/OFF
SPINDL/OFF
FINI
```

A single \$ at the end of a line continues the statement on the next line.



Reading the motion

GO/T0, L1, T0, PL1, T0, L4 — touch L1 and L4 at depth PL1.
TLRGT — tool on the right of the drive line (outside).
PAST — outside corner; TANTO — change surface at the tangent point.

Manual Programming vs. APT

Aspect	Manual (G-code)	APT
Calculations	by the programmer (tangent points, offsets)	by the computer
Language	numbers and codes	English-like words
Machine-specific	yes	no — post-processor adapts it
Cutter offset	calculated or G41/G42	automatic from CUTTER/
Best for	simple 2-D, drilling	complex contours, 3- to 5-axis

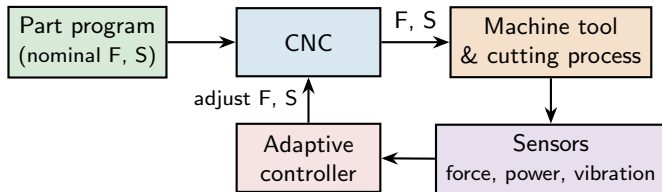
Key idea

APT introduced the ideas that CAD/CAM systems still use today: geometry first, then tool motion, then a post-processor for each machine.

What is Adaptive Control?

Adaptive control (AC)

A control system that **measures** process variables during machining (force, torque, power, temperature, vibration) and **automatically adjusts** feed and speed to suit the actual cutting conditions.



- Ordinary NC uses the programmed feed even if the depth of cut or hardness changes.
- The programmer must choose *safe* (slow) values for the worst case — AC removes that penalty.

Types of Adaptive Control

ACC — Constraint

- Keep a variable (force, power) **at or below a limit**.
- Feed increased in light cuts, reduced in heavy cuts.
- Most common in practice.

ACO — Optimisation

- Maximise a **performance index** — e.g. metal-removal rate per cost.
- Needs on-line tool-wear sensing.
- Mostly research systems.

GAC — Geometric

- Measure the part (probe) and correct for tool wear & deflection.
- Used in finishing cuts.
- Improves **accuracy**.

Benefits

Higher metal-removal rate, fewer broken tools, protection of spindle and part, less programmer effort (no need to guess conservative feeds), consistent quality.

Adaptive Control Constraint: Worked Example

Problem: In a milling cut the cutting force is approximately

$$F_c = k d f$$

(d = depth of cut, f = feed per tooth). With $d = 2$ mm and $f = 0.3$ mm the force equals the limit $F_{\max} = 1500$ N. The depth suddenly rises to 3 mm. What feed must the ACC system select?

Solution:

$$k = \frac{1500}{2 \times 0.3} = 2500 \text{ N/mm}^2$$

$$F_c \text{ (no AC)} = 2500 \times 3 \times 0.3 = 2250 \text{ N (overload!)}$$

$$f_{\text{new}} = \frac{1500}{2500 \times 3} = \boxed{0.2 \text{ mm}}$$

When the cut becomes lighter again, the ACC raises the feed back up — the force stays at the limit all the time.

Advantages and Disadvantages of NC

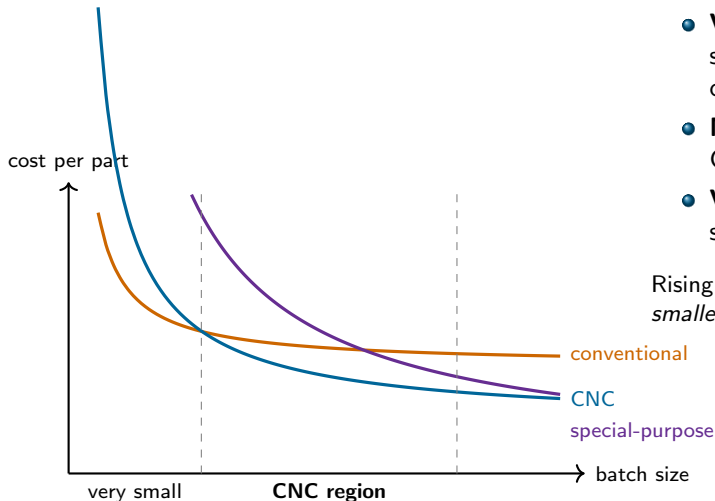
Advantages

- Less non-productive time (set-up, handling, measuring).
- Greater accuracy & repeatability, less scrap.
- Complex shapes are possible.
- Fewer jigs & fixtures, smaller inventory.
- Quick design changes (edit the program).
- Shorter manufacturing lead time.

Disadvantages

- High investment cost.
- Higher maintenance effort & skilled technicians.
- Part programming effort and training.
- Higher machine-hour rate — needs high utilisation (2–3 shifts).

Where NC Pays: Cost vs. Batch Size



- **Very small batches** — programming & set-up dominate; conventional machines can be cheaper.
- **Medium batches, many part types** — CNC is cheapest.
- **Very large volumes of one part** — special-purpose / transfer machines win.

Rising part complexity pushes the CNC region to *smaller* batches.

Cost Elements and Justification Methods

Fixed (investment) costs

Machine & controller, installation, tooling, software (CAM), training.

Operating costs

Operator labour, programming, maintenance, power, tool wear, overheads — usually combined into a **machine-hour rate**.

Justification methods

Payback period = $\frac{\text{investment}}{\text{net annual saving}}$

Break-even quantity: batch size where both methods cost the same.

Return on investment / net present value (for larger projects).

Example: a CNC costs Rs 40 lakh and saves Rs 10 lakh per year $\Rightarrow$ payback = $40/10 =$ 4 years.

Break-Even Analysis: Worked Example

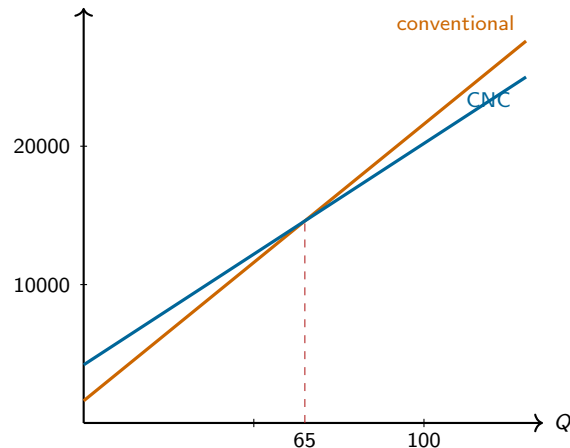
	Conventional	CNC
Machine-hour rate	Rs 400/h	Rs 1200/h
Set-up per batch	4 h	1 h
Programming / batch	—	Rs 3000
Cycle time / part	30 min	8 min

$$C_{\text{conv}} = 400 (4 + 0.5 Q) = 1600 + 200 Q$$

$$C_{\text{CNC}} = 3000 + 1200 (1 + \frac{8}{60} Q) = 4200 + 160 Q$$

$$1600 + 200 Q = 4200 + 160 Q \Rightarrow \boxed{Q = 65 \text{ parts}}$$

batch cost (Rs)



Batches **above 65** parts are cheaper on the CNC;
smaller batches on the conventional machine.

Summary & What Comes Next

Concept

- Automation: elements, types, USA principle, 10 strategies.
- Improved devices & manipulators on conventional machines.
- NC principles: axes, PTP / contouring, open / closed loop, BLU, interpolation.
- CNC, DNC and machining centres.
- Coding: tape codes, block formats, G- & M-codes.
- APT programming, adaptive control, economics.

Practice makes perfect

- Write G-code for a new part and check it in a free simulator.
- Re-work the BLU, ACC and break-even examples with new numbers.
- Run the interpolation C program for other end points.

Coming up

CAD/CAM, industrial robotics and flexible manufacturing systems.

Thank You

Questions?